

Stress in ion exchanged soda-lime silicate and sodium aluminosilicate glasses: Experimental and theoretical comparison

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Abstract

Glass samples of two different chemical compositions were strengthened by ion exchange. Residual stress profiles were experimentally determined by differential surface refractometry. The experimental results were compared with values theoretically predicted by a formalized viscoelastic mathematical model incorporating both stress buildup and stress relaxation contributions. The comparison exhibited a remarkable agreement both in terms of strengthening characteristics and stress profiles curves between experimentally measured and theoretically predicted values. The mathematical model incorporated terms directly linked to selected relaxation mechanisms either isochoric shear stress driven and fast relaxation. The comparison demonstrated the relevance of the stress relaxation mechanisms for the different glass chemical compositions. The approach presented in this study will allow the prediction of the mechanical behavior of chemically strengthened glass starting from its chemical composition together with the ion exchange process parameters.

KEYWORDS

chemical strengthening, glass, ion exchange, residual stress, stress relaxation

1 | INTRODUCTION

Chemical strengthening of glass by ion exchange is a process which is performed below the glass transition temperature and it is recording a growing technological relevance in the application of high-strength structural glazing.^{1,2,3,4} Glass strength is determined by mechanical testing of samples up to their ultimate failure by breakage. Glass samples are tested in well-defined load and constraints conditions so that stress can be readily evaluated. The value of stress at breakage resulting from test conditions is the strength value. As pointed out by Macrelli,⁵ prediction of strength, for chemically strengthened glass, is somehow problematic. For chemically strengthened glass by ion exchange it cannot be simply assumed that final

strength is the sum of original strength and of the induced surface compression as for thermally strengthened glass.⁵ This statement is acceptable provided that the depth of the compression layer is much larger than the maximum magnitude of the depth of surface flaws. To correctly predict the final strength, both the original strength before strengthening and the residual stress profile caused by the strengthening process should be known.⁵ Prediction of residual stress profile generated by ion exchange is still a matter of scientific interest.^{5,6,7} Therefore, several approaches have been proposed in the literature as either experimental^{8,9} or theoretical.^{10,11} Despite some critical issues highlighted in the literature^{8,9,10} the experimental method based on differential surface refractometry^{12,13} is becoming widely used in industrial applications.¹⁴ This method is used for the determination of surface

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compression, that is residual stress value at the surface, and compression layer depth, that is the distance from the surface to the inner point of the glass where the residual stress is zero changing from compressive to tensile state. In some instrumental implementations it is also possible to reconstruct the residual stress profile.^{9,15} Theoretical prediction of residual stress profile is based on the introduction of an incompatible strain generated by the ions' invasion of the glass structure in an analogy with thermoelasticity.^{11,16,17} A full mathematical model of residual stress generated in ion exchange processes is made problematic by the presence of different relaxation mechanisms in the glass structure. Recently, a mathematical model has been proposed by Macrelli, Varshneya, and Mauro¹¹ for a complex residual stress profile resulting from the combination of different relaxation mechanisms in lithium magnesium aluminosilicate glasses submitted to ion

exchange of lithium ions in the glass for sodium ions provided by molten salt of sodium nitrate.

The purpose of this study is to present, in the first part, experimental results of residual stress profile for soda-lime silicate glass (TS11) and sodium aluminosilicate glass (AS11) submitted to ion exchange of sodium ions in the glass structure for potassium ions provided by the molten salt of potassium nitrate. This study is focused on TS11 and AS11 glass chemical compositions because they are the most used for glass chemical strengthening, whereas sodium aluminosilicates are mostly used in consumer electronics and transportation and soda-lime silicates are used in architectural and transportation.

In the second part of this study a mathematical model is presented based on the one introduced by Macrelli, Varshneya, and Mauro.¹¹ The model, suitably adapted for the

TABLE 1 List of symbols and definitions

Symbol	Units	Definition
σ_{EB}	MPa	Residual equi-biaxial stress
Sc	MPa	Surface Compression – Value of residual stress at glass surface ($Sc = \sigma_{EB}(0)$)
Cd	μm	Compression layer depth – Depth at which residual stress is zero. $\sigma_{EB}(Cd) = 0$
ρ	Kg/m^3	Glass Density
x	μm	Spatial Coordinate
t	s	Time
D	$\mu m^2/s$	Interdiffusion coefficient
$C(x,t)$	mol/m^3	Ion concentration
C_o	mol/m^3	Initial concentration of ions in the glass to be exchanged
C_s	mol/m^3	Equilibrium surface concentration of incoming ions.
$c(x,t)$	Adimensional	Relative concentration $c(x,t) = C(x,t)/C_o$
χ	Adimensional	Relative equilibrium surface concentration – $\chi = C_s/C_o$
B	mol^{-1}	Linear Network Dilation Coefficient (Cooper Coefficient)
E	MPa	Young modulus
ν	Adimensional	Poisson ratio
$V(x,t)$	Adimensional	Varshneya Factor ¹¹
V_o	Adimensional	Constant term of the Varshneya Factor
$R(t)$	Adimensional	Relaxation function
τ	s	Relaxation time of the relaxation function
b	Adimensional	Exponent of the non-Maxwellian stretched relaxation function
ψ	Adimensional	Coefficient of the linear part of the Varshneya factor
n_{TM}	Adimensional	Refractive index – TM component
n_{TE}	Adimensional	Refractive index – TE component
Γ	MPa^{-1}	Stress Optical Coefficient

TABLE 2 Glass chemical compositions (mol%)

	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	TiO ₂	CaO	MgO	Na ₂ O	K ₂ O	SO ₃	B ₂ O ₃
TS11	70.98	0.57	0.03	0.04	8.87	6.44	12.85	0.04	0.15	—
AS11	66.00	11.20	0.01	0.01	0.31	6.18	13.26	2.89	0.02	0.11

specific glass compositions herewith considered, is applied to the glasses and ion exchange parameters described in the experimental part. An additional purpose of this study is to demonstrate how the general model can be applied to a wide class of glasses just by selecting the most appropriate relaxation parameters. Finally, theoretically predicted values were compared with the experimental ones. A list of symbols with their units and definitions used in this study is reported in Table 1.

2 | EXPERIMENTAL PART

Commercial soda-lime silicate (Şişecam Flat Glass, Turkey) and sodium aluminosilicate (Schott AS 87 for Xensation®) glass samples with the dimensions of 60 mm × 60 mm × 1.1 mm were used in the study. Glass compositions were determined by using X-ray fluorescence spectroscopy and referred as TS11 and AS11 respectively (see Table 2). The glass transition temperatures were determined by dilatometer analysis (DIL402 PC, Netzsch, Germany) and densities of glasses ρ (kg/m³) were measured at room temperature by the Archimedes' method with density kits by using Mettler Toledo XS204 instrument. The measurements were carried out according to ASTM C693 standard. The glass transition and density values were reported listed in Table 3.

Before ion exchange treatments, the samples were ultrasonically cleaned in deionized water and isopropyl alcohol for 5 minutes, sequentially. After cleaning, the glass samples were treated with dry nitrogen gas.

Lema TC 804A furnace (Parma, Italy) was used for ion exchange treatments and 20 samples were used for each process condition. In order to prevent thermal shock, samples were held above the salt bath for 60 minutes before submersion into the molten KNO₃ (99.0% purity, Quality Chemicals, Spain) salt bath. Salt bath temperature and immersion time were used as main process parameters. Salt

bath temperatures for both TS11 and AS11 glasses were selected as 400, 425, and 450°C. Immersion times of 12, 24, and 36 hours were used for TS11 glasses while 1, 2, 4, and 8 hours of immersion time were used for AS11 glasses. After the ion exchange treatments, sample cleaning procedure was repeated using deionized water and dry nitrogen gas.

The compressive biaxial stress (σ_c) and depth of layer (C_d) were determined by using FSM6000 LE (Luceo Co Ltd.) instrument.¹⁵ The working principle of such instruments is based on interference effects (fringes) generated by the waveguide nature of ion exchanged glass due to refractive index gradients introduced by the ion exchange process.^{14,15} Stress was determined considering birefringence between the transverse electric (TE) and transverse magnetic (TM) light propagation modes. The measurements were made on five different areas of four samples for each treatment according to ASTM C1422 standard.¹⁴ C_d was determined from the point where residual stress becomes zero, which corresponds to the coordinate where refractive index of TE and that of TM modes merge to a single value. In order to have accurate C_d measurements, the stress optical coefficient and refractive index were given as input to measurement software. Stress optical coefficient values were taken from literature and technical data sheet. For TS11 glasses, stress optical coefficient has been considered as 2.83 TPa⁻¹, and for AS11 glasses it was taken as 2.90 TPa⁻¹. Refractive index values for both glass types were directly measured using the Schmidt-Haensch refractometer. Refractive index for TS11 glasses in the bulk (not exchanged) part resulted in 1.52 and for AS11 glasses resulted in 1.51. Residual stress profiles were determined accordingly based on the hypothesis, introduced by Kishii,¹² of linear refractive index distribution. This last hypothesis is somehow questionable, and it is a matter of scientific discussion,¹⁰ nevertheless it can be considered an acceptable initial working assumption.

TABLE 3 Physical characteristics of the glasses

	ρ kg/m ³	T_g °C	V_m^* m ³ /kmol	$V_m\text{-IX}^*$ m ³ /kmol	$E\text{-IX}^*$ GPa	ΔV_m m ³ /kmol	$B\cdot C$	B mol ⁻¹
TS11	2486	560.0	0.02362	0.02539	72.27	0.00177	0.024979	0.001944
AS11	2448	627.6	0.02635	0.02818	71.02	0.00183	0.023163	0.001747

Note: ρ —Density (kg/m³).

T_g —Glass transition temperature (°C).

V_m —Molar Volume (m³/kmol); $V_m\text{-IX}$ —Molar Volume after ion exchange (m³/kmol).

$E\text{-IX}$ —Young Modulus after Ion Exchange (GPa).

ΔV_m —Change in molar volume due to Ion Exchange (m³/kmol).

$B\cdot C$ —Incompatible strain induced by Ion Exchange.

$B = \frac{1}{3} \frac{\partial V_m}{\partial C} \approx \frac{1}{3} \frac{\Delta V_m}{V_m - C_0}$ - Linear Network Dilation Coefficient LNDC (Cooper Coefficient) (mol⁻¹).

*Determined by Sci-Glass 6.5.

3 | THEORETICAL PART

The mathematical model of residual stress profile considered in this study is the one introduced by Macrelli, Varshneya, and Mauro.¹¹ This model was originally developed to simulate the glass network evolution during chemical strengthening by ion exchange. In particular, the mathematical model had the purpose to justify the formation of a subsurface compression maximum and to predict the reversal of compression stress into tensile stress for a certain class of silicate glasses (lithium magnesium aluminosilicate glass) upon prolonged ion exchange. In order to reproduce experimental results, the model incorporated different relaxation mechanisms as isochoric (shear stress driven) and non-isochoric (hydrostatic stress driven) and the structural relaxation effect was driven by free-energy (topologically driven). One of the purposes of this study is to evaluate how this general model shall be reduced to reproduce acceptable predictions of equi-biaxial residual stress in two other chemical compositions of ion exchanged glasses. The model requires an expression for the concentration profile of the incoming ions and a number of parameters, some of them measured (chemical composition and density of the investigated glasses) and others (Young modulus, Poisson ratio, Linear network dilation coefficient—LNDC also known as B-Cooper coefficient) evaluated by knowing the chemical composition.¹⁸ Other parameters like relaxation time τ for the relaxation function, interdiffusion coefficient D , and equilibrium surface concentration ratio χ were let free for the purpose of best fit data. The concentration profile of the incoming ions was assumed to follow the complementary error function which is the solution to the diffusion equation for a semi-infinite medium with constant surface boundary condition (Constant surface concentration attained almost instantaneously at the beginning of the immersion time)^{11,17}:

$$C(x,t) = C_s \operatorname{erfc} \left(\frac{x}{2\sqrt{Dt}} \right), \quad (1)$$

where “ $\operatorname{erfc}(z)$ ” is the complementary error function¹⁷ and C_s is the incoming ions concentration at glass surface. Let's indicate with C_o the concentration of ions originally present in the glass and to be exchanged for ions coming from the external source. From this definition it comes that $C_s \leq C_o$ and $C_s = C_o$ only if all available ions in the surface are exchanged. Equation (1) can be rewritten in terms of relative concentration $c(x,t) = C(x,t)/C_o$:

$$c(x,t) = \frac{C(x,t)}{C_o} = \chi \operatorname{erfc} \left(\frac{x}{2\sqrt{Dt}} \right), \quad (2)$$

where $\chi = C_s/C_o$ is the equilibrium value of surface concentration, that is the fraction of exchanged ions at the glass surface.

From concentration distribution the residual stress equation is derived¹¹ for equi-biaxial stress $\sigma_{EB}(x,t)$ as follows:

$$\sigma_{EB}(x,t) = -\frac{BE}{1-\nu} \left[\left[V(x,t)C(x,t) - \overline{VC(t)} \right] - \int_0^t \frac{\partial R(t-\theta)}{\partial \theta} \left[V(x,\theta)C(x,\theta) - \overline{VC(\theta)} \right] d\theta \right] \quad (3)$$

where E is the elastic Young modulus of the glass, ν is the Poisson ratio, and B is the linear network dilation (LNDC) coefficient also named Cooper coefficient^{16,17}. $R(t)$ is the viscous relaxation function that, for this study, we will express with a non-Maxwellian stretched relaxation function¹⁷:

$$R(t) = \exp \left[-\left(\frac{t}{\tau} \right)^b \right], \quad (4)$$

where τ is the relaxation time and b is the stretched exponential ($b = 3/7$) exponent.

$V(x,t)$ is a suitable function named “Varshneya-Factor”.¹¹ The purpose of introducing this function is to incorporate the main relaxation contributions: fast β , slow α , and structural relaxations.

Equation (3) presents two terms: the first term is the stress buildup one while the second is the relaxation term which includes stress relaxation effects. When applying the mathematical model to the glass considered in this study (soda-lime and sodium aluminosilicate) we will make the assumption that there is practically no effect due to free-energy structural relaxation and the only remarkable relaxation effects are^{19,20} the fast β -relaxation and the slow isochoric, shear stress driven, α -relaxation effects. In Binder and Kob¹⁹ terminologies “fast β -relaxation” and “slow α -relaxation” used in this study were clarified. This is also to avoid confusion with other β -processes better known in the literature as “Johari-Goldstein β -process” or “slow β -process” that is found in several glass formers. As reported in Binder and Kob,¹⁹ (see Figure 5.3 and Figure 5.4 of reference [19]) the fast β -relaxation and slow α -relaxation can be identified looking to the correlation function vs time where, for relaxation

TABLE 4 Constant parameters used in the mathematical model—Glass AS11

T (s)	E(MPa)	ν	BC	V_o	Ψ
1×10^7	71 020	0.23	0.023163	0.63	0.10

Temperatures and parameters		Ion exchange times			
Temperature	Parameter	1 h	2 h	4 h	8 h
400°C	D ($\mu\text{m}^2/\text{s}$)	0.0060	0.0060	0.0060	0.0060
	χ	0.80	0.83	0.90	0.90
425°C	D ($\mu\text{m}^2/\text{s}$)	0.0120	0.0120	0.0120	0.0120
	χ	0.85	0.90	0.93	0.95
450°C	D ($\mu\text{m}^2/\text{s}$)	0.0195	0.0195	0.0195	0.0195
	χ	0.87	0.90	0.93	0.95

TABLE 5 Variable parameters used in the mathematical model—Glass AS11**TABLE 6** Constant parameters used in the mathematical model—Glass TS11

E(MPa)	ν	BC	V_0	χ
72 270	0.23	0.02498	0.65	0.90

phenomena below T_g , they can be identified after the “boson peak”. Considering stress relaxation dynamics they can be identified:

- A first fast β -stress relaxation. During this step the alkali ions jump from one alkali ion cage into another with a characteristic time period (relaxation time) from few ps to few ns. There is no change of Si–O network topology during this step.
- A second slow α -stress relaxation. During this step the Si–O network (short range glass structure) topology is modified, this occurs after several ns and may be prolonged several hours or days. As evident there is overlap between the two relaxation stages.

Temperatures and parameters		Ion exchange times		
Temperature	Parameter	12 h	24 h	36 h
400°C	$\tau(\text{s})$	4.0×10^6	4.0×10^6	4.0×10^6
	D ($\mu\text{m}^2/\text{s}$)	1.40×10^{-4}	1.40×10^{-4}	1.40×10^{-4}
	Ψ	0.32	0.32	0.32
425°C	$\tau(\text{s})$	1.94×10^6	1.94×10^6	1.94×10^6
	D ($\mu\text{m}^2/\text{s}$)	3.40×10^{-4}	3.40×10^{-4}	3.40×10^{-4}
	Ψ	0.32	0.32	0.32
450°C	$\tau(\text{s})$	1.00×10^6	1.00×10^6	1.00×10^6
	D ($\mu\text{m}^2/\text{s}$)	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
	Ψ	0.37	0.37	0.35

TABLE 7 Variable parameters used in the mathematical model—Glass TS11

Temperatures (T) and parameters (P)		Ion exchange times							
T	P	1 h		2 h		4 h		8 h	
		Exp.	Prd.	Exp.	Prd.	Exp.	Prd.	Exp.	Prd.
400°C	Cd	16	17	22	23	29	31	40	42
	Sc	905	900	925	914	943	957	946	929
425°C	Cd	22	23	29	31	39	42	56	55
	Sc	956	943	982	972	998	975	984	960
450°C	Cd	27	29	37	38	49	51	69	67
	Sc	957	958	952	967	944	967	910	949

TABLE 8 Residual Stress parameters: Cd(μm); Sc (MPa)—Glass AS11

Note: Comparison between experimental (Exp.) and theoretically predicted (Prd.) values. Surface Compression (Sc) Comp. Layer Depth (Cd).

TABLE 9 Residual Stress parameters: Cd(μm); Sc (MPa)—Glass TS11

Temperatures (T) and parameters (P)		Ion exchange times					
		12 h		24 h		36 h	
		Exp.	Prd.	Exp.	Prd.	Exp.	Prd.
400°C	Cd	7	10	12	14	13	16
	Sc	647	657	639	624	619	600
425°C	Cd	12	15	17	20	20	24
	Sc	618	621	577	578	548	548
450°C	Cd	16	19	23	26	28	31
	Sc	508	508	465	461	454	454

Note: Comparison between experimental (Exp.) and theoretically predicted (Prd.) values.

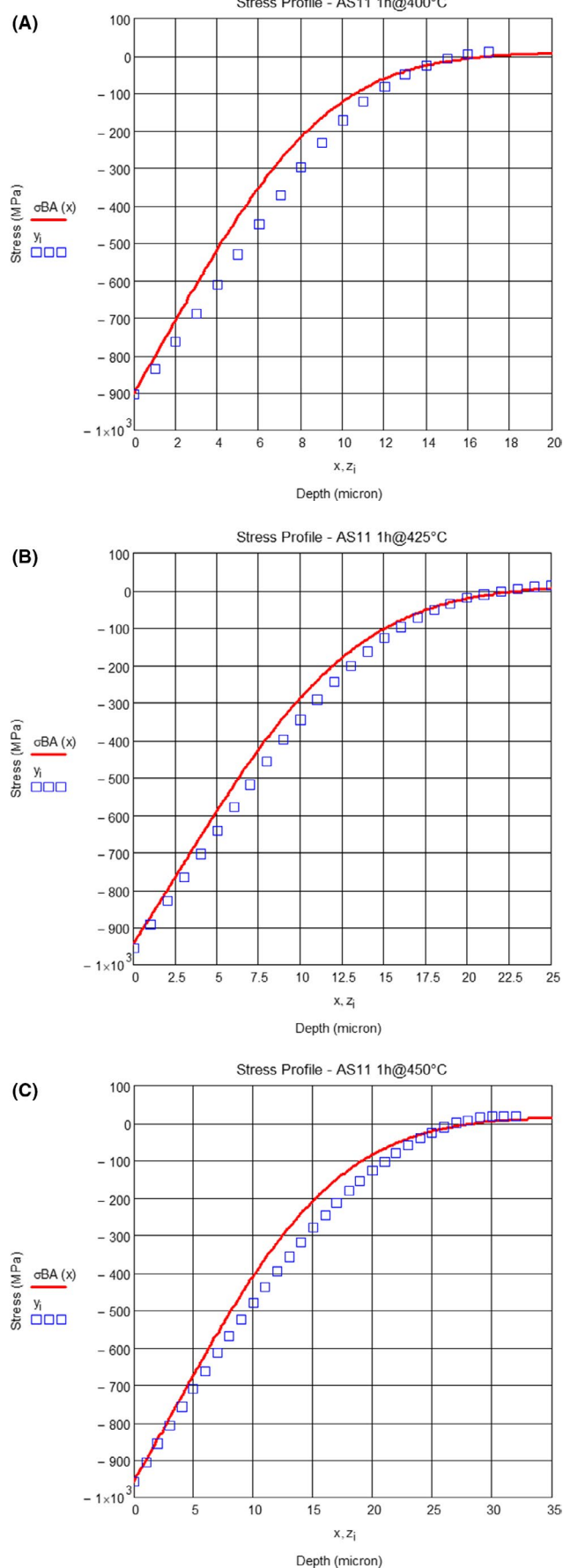
Surface Compression (Sc) Comp. Layer Depth (Cd).

These two contributions can be expressed by a constant factor V_0 for fast β and by a linearly concentration depending term for slow α through another constant factor Ψ :

$$V(x,t) = V_0 - \Psi \cdot c(x,t), \quad (5)$$

In¹¹ $\Psi(t)$ was proposed through a non-Maxwellian stretched exponential function. In the present study we kept $\Psi(t)$ a constant parameter Ψ as we are not expecting a structural free energy-driven relaxation mechanism resulting eventually in a stress reversal from compressive to tensile. Numerical calculations were performed by using MathCad 15.0 (Parametric Technology Corporation - PTC Inc). The computational parameters considered for AS11 glass and TS11 glass are reported in Tables 4-7 respectively. The fitting strategy followed in the present study is to get the best possible matching between experimentally measured and theoretically predicted values of surface compression and compression layer depth. Experimentally determined values of surface compression were considered for the determination of the constant parameter V_0 . Constraints in the numerical selection were in the range of V_0 ($0 < V_0 < 1$). The

FIGURE 1 A, Comparison between measured and predicted stress profile—Glass AS11, 1 h immersion time at 400°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. B, Comparison between measured and predicted stress profile—Glass AS11, 1 h immersion time at 425°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. C, Comparison between measured and predicted stress profile—Glass AS11, 1 h immersion time at 450°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values [Color figure can be viewed at wileyonlinelibrary.com]



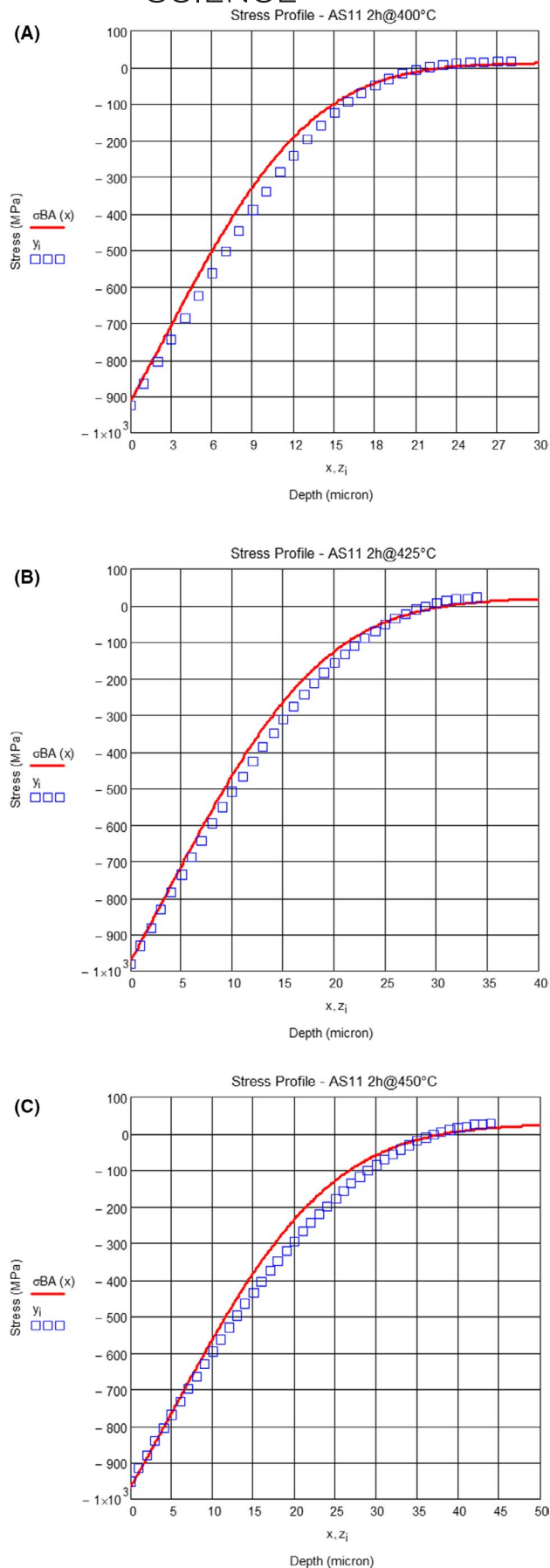


FIGURE 2 A, Comparison between measured and predicted stress profile—Glass AS11, 2 h immersion time at 400°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. B, Comparison between measured and predicted stress profile—Glass AS11, 2 h immersion time at 425°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. C, Comparison between measured and predicted stress profile—Glass AS11, 2 h immersion time at 450°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values [Color figure can be viewed at wileyonlinelibrary.com]

strategy used in the determinations of the parameters of the mathematical model has been driven by considerations related to the physical nature of the involved parameters that can be grouped in parameters related to interdiffusion kinetics: D and χ and parameters related to stress relaxation: V_0 , Ψ , τ . Interdiffusion coefficient D is clearly depending from the temperature of the ion exchange process by an Arrhenius type function through an activation energy value. In general D can also be considered depending on the time of ion exchange but, in this study, we have taken the assumption that it is only depending upon temperature, considering that the time dependence is a second order effect. As for the equilibrium surface relative concentration χ , looking to surface compression data and also based on additional, yet not published, concentration depth profile data, we have found that the achievement of an equilibrium concentration is a fast process for soda-lime glass (TS11) not significantly depending on temperature, for this reason it has been taken constant for TS11 glass. For sodium aluminosilicate glass it has been found that the above condition is no more valid and the equilibrium value is a temperature- and time-dependent function. For this reason, we have left it as a variable for AS11 glass. With respect to stress relaxation parameters, V_0 is related to fast β -relaxation mechanism. As discussed above its typical relaxation time is the order of “ps” to “ns” that can be considered almost instantaneous when compared to ion exchange times, for this reason it is maintained constant for both AS11 and TS11 glass and its value was determined to match surface compression values. Ψ and τ parameters are related to isochoric α -relaxation mechanism and their variability is related to the connectivity of the Si—O network and to the difference from ion exchange temperature and glass transition temperature. Sodium aluminosilicate glass has a better network connectivity and a higher T_g (627.6°C) when compared to soda-lime glass ($T_g = 560^\circ\text{C}$). For these reasons we kept them fixed for AS11 and variable for TS11.

4 | RESULTS

The main residual stress parameters, S_c (MPa) and C_d (μm) were reported in Tables 8 and 9. A comparison was presented

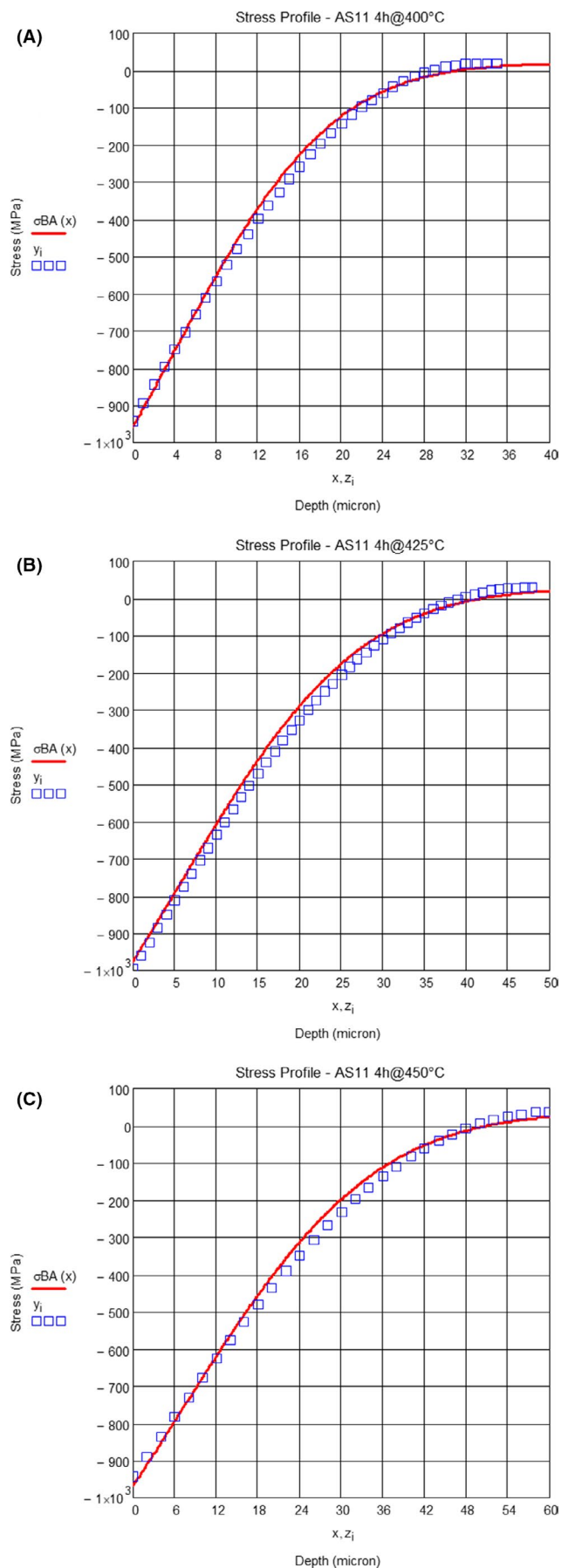


FIGURE 3 A, Comparison between measured and predicted stress profile—Glass AS11, 4 h immersion time at 400°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. B, Comparison between measured and predicted stress profile—Glass AS11, 4 h immersion time at 425°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. C, Comparison between measured and predicted stress profile—Glass AS11, 4 h immersion time at 450°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values [Color figure can be viewed at wileyonlinelibrary.com]

between experimentally determined values with the ones predicted theoretically using the above described formalized viscoelastic mathematical model. The accuracy of the experimental values is declared by the instrument manufacturer¹⁵ for Sc and Cd values in ± 20 MPa and ± 5 μm , respectively. Results of the measured residual stress profiles compared with the corresponding results of the theoretically predicted values are reported in Figures 1-4 for AS11 glass and in Figures 5-7 for TS11 glass, respectively. In case of AS11 glass, results are grouped in each figure number by the same immersion time: 1 hour immersion time 400°C (Figure 1A), 425°C (Figure 1B), and 450°C (Figure 1C); 2 hours immersion time 400°C (Figure 2A), 425°C (Figure 2B), and 450°C (Figure 2C); 4 hours immersion time 400°C (Figure 3A), 425°C (Figure 3B), and 450°C (Figure 3C); 8 hours immersion time 400°C (Figure 4A), 425°C (Figure 4B), and 450°C (Figure 4C). For TS11 glass results are grouped in each figure number by the same ion exchange temperature: 400°C (Figure 5A) 12 hours immersion time, (Figure 5B) 24 hours immersion time, and (Figure 5C) 36 hours immersion time; 425°C (Figure 6A) 12 hours immersion time, (Figure 6B) 24 hours immersion time, and (Figure 6C) 36 hours immersion time; 450°C (Figure 7A) 12 hours immersion time, (Figure 7B) 24 hours immersion time, and (Figure 7C) 36 hours immersion time.

The reason of the different grouping for the two different glass types is related to the point that sodium aluminosilicate glass has a negligible stress relaxation and stress profile changes are mostly driven by changes in the interdiffusion kinetics parameters. For this reason we have chosen to group data at constant immersion time for different temperatures. Soda-lime glass on the contrary presents a significant stress relaxation, for this reason we have chosen to group them at constant temperature to highlight changes of stress profile due to different immersion times at same temperature.

The predicted residual stress profile values are represented by continuous red lines while the experimentally determined values are represented by blue boxes. The calculated values have been determined with a spatial step of 1 μm by a direct calculation of Equation (3) using the parameters

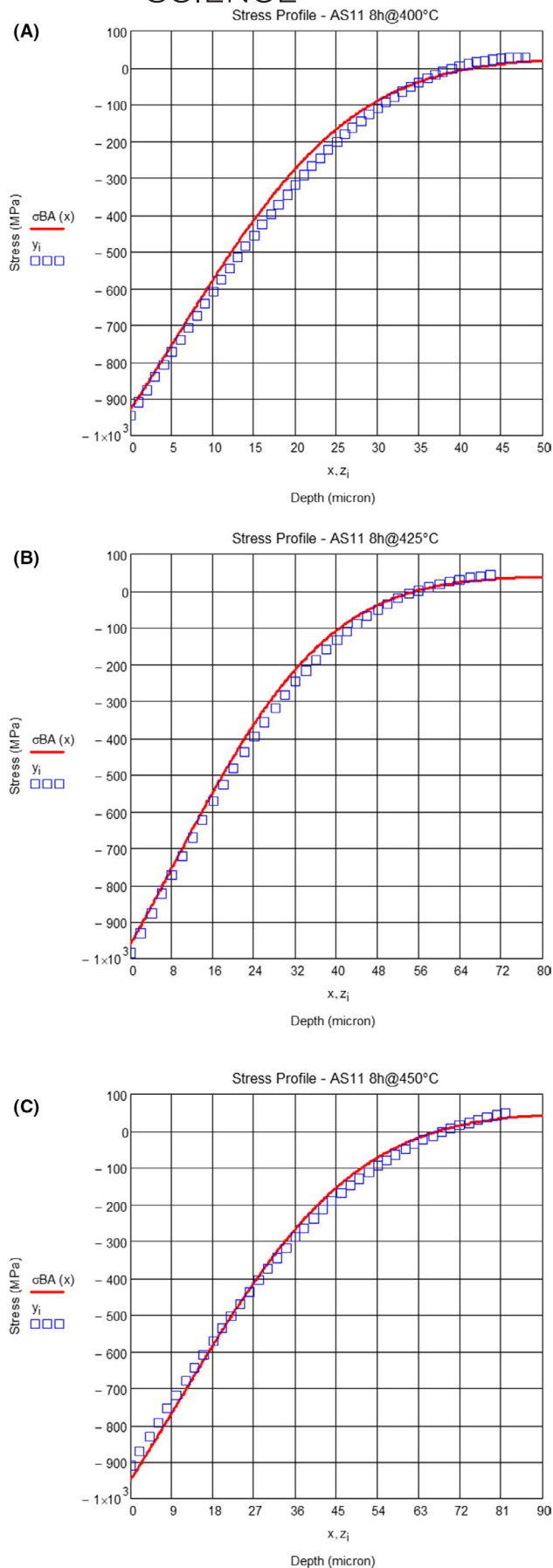


FIGURE 4 A, Comparison between measured and predicted stress profile—Glass AS11, 8 h immersion time at 400°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. B, Comparison between measured and predicted stress profile—Glass AS11, 8 h immersion time at 425°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. C, Comparison between measured and predicted stress profile—Glass AS11, 8 h immersion time at 450°C. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values [Color figure can be viewed at wileyonlinelibrary.com]

of Tables 4-7. The experimentally measured values have been determined on the hypothesis of a linear gradient of the refractive index modes TE and TM.

5 | DISCUSSION

The experimentally and theoretically determined Sc and Cd values reported in Tables 8 and 9 are quite comparable for both types of glasses within the accuracy limits declared by the instrument manufacturer. The results reported in Figures 1-7 related to the complete residual stress profiles demonstrate the capability of the proposed mathematical model to predict residual stress values for two different types of silicate glasses chemical compositions and for ion exchange processes on a wide range of temperatures and immersion times. The number of variable parameters has been taken as limited as possible to those expected to depend on temperature (interdiffusion coefficient D , relaxation time τ and surface equilibrium concentration χ) and immersion time. Other approaches have been recently proposed by Sun and Dugnani²¹ for stress relaxation modeling of ion exchanged soda-lime glass based on Prony series for the non-isochoric relaxation function, concentration-dependent relaxation times, and concentration-dependent Cooper coefficient. We have intentionally limited the number of fitting parameters to highlight the effects of the different relaxation mechanisms.

Equi-biaxial stress profile measurements performed by Differential Surface Refractometry (DSR)^{14,15} deserve a particular discussion. The DSR technique implemented in this FSM 6000 instrument provides an acceptable estimate of surface compression (Sc) and compression layer depth (Cd) for soda-lime silicate and sodium aluminosilicate glasses. The measurement is based on the waveguide effect generated in the near surface layer of the glass subjected to ion exchange by the refractive index gradient induced by the hosted ions with larger molar volume and polarizability. The stress distribution in a glass after ion exchange on all its surfaces is equi-biaxial.¹¹ This is due to the condition of free deformation along the diffusion direction coupled

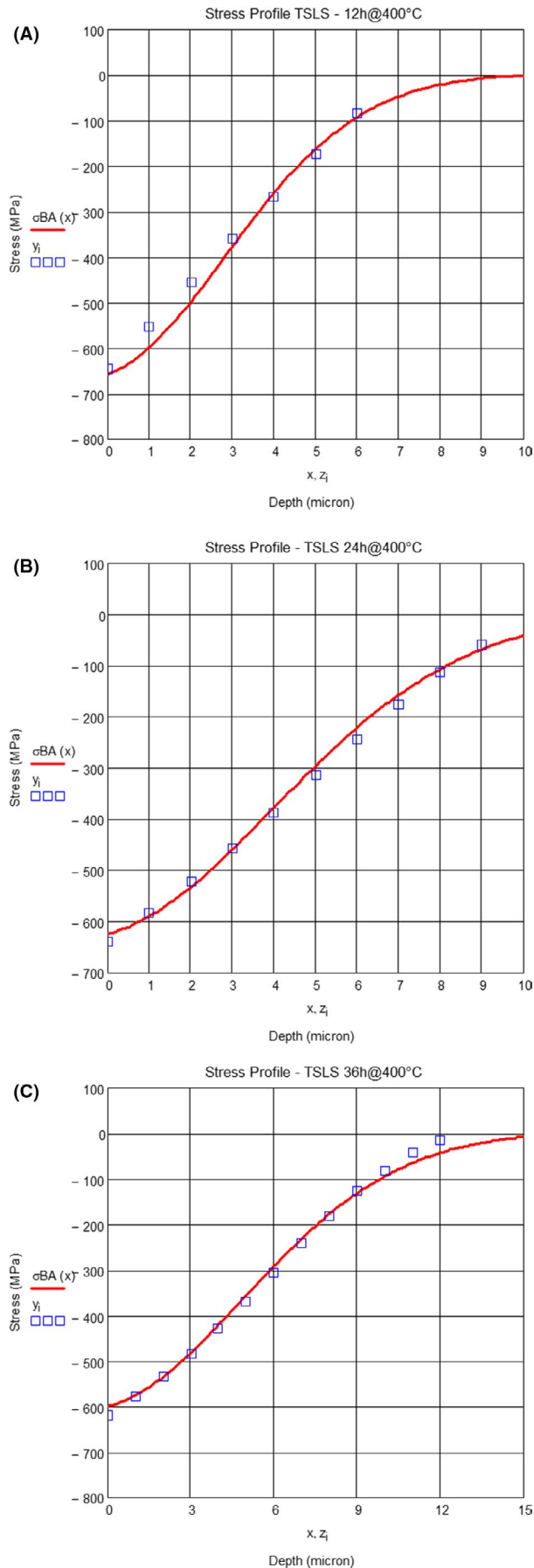


FIGURE 5 A, Comparison between measured and predicted stress profile—Glass TS11 at 400°C, immersion times 12 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. B, Comparison between measured and predicted stress profile—Glass TS11 at 400°C, immersion time 24 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. C, Comparison between measured and predicted stress profile—Glass TS11 at 400°C, immersion time 36 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values [Color figure can be viewed at wileyonlinelibrary.com]

with the constrained strain condition along the two other spatial directions. When a monochromatic light beam, split into its transverse magnetic (TM) and transverse electric (TE) polarization directions, enters into a stressed glass layer, optical birefringency effects allow stress determination by knowing the relevant stress optical coefficient. Both components (either TM or TE) of the light beam are bent by the refractive index gradient up to the point where they are turned up toward the surface by total reflection.¹² This last condition determines interference fringes for both TM and TE components. By analyzing these two systems of interference fringes the depth of the turning point can be determined.¹² The depth of the turning point is believed to be a good estimate of the compression layer depth although, by definition, it is always slightly lower. In optical terms, the exact definition of compression layer depth, is given in correspondence to the x coordinate where the refractive indices of the two components TM and TE merge in a single value.

In this study for the determination of the equi-biaxial stress, we assumed a linear refractive index gradient for both TM and TE components. Equi-biaxial stress at a depth “ x ” from the glass surface has been evaluated according to:

$$\sigma_{EB}(x) = \frac{n_{TM}(x) - n_{TE}(x)}{\Gamma}, \quad (6)$$

where Γ is the stress optic coefficient assumed independent of both chemical composition (which is indeed altered by ion exchange) and stress itself at point x . The assumption of linear refractive index gradient is a first-order assumption that may lead to even significant uncertainty contribution to the determination of residual stress profiles.⁹ The agreement of predicted to measured values is remarkable for AS11 glass (Figure 1 to Figure 4). A similar level of agreement can be noticed also for TS11 glass (Figure 5 and Figure 6) while a less accuracy can be noticed in Figure 7 for ion exchange processes at higher temperature and when immersion time is 24 and 36 hours. Comparing Figure 7 to Figure 5 and Figure 6 it can be concluded that the disagreement is a temperature-related issue more than a time immersion one. The disagreement in the predicted and measured

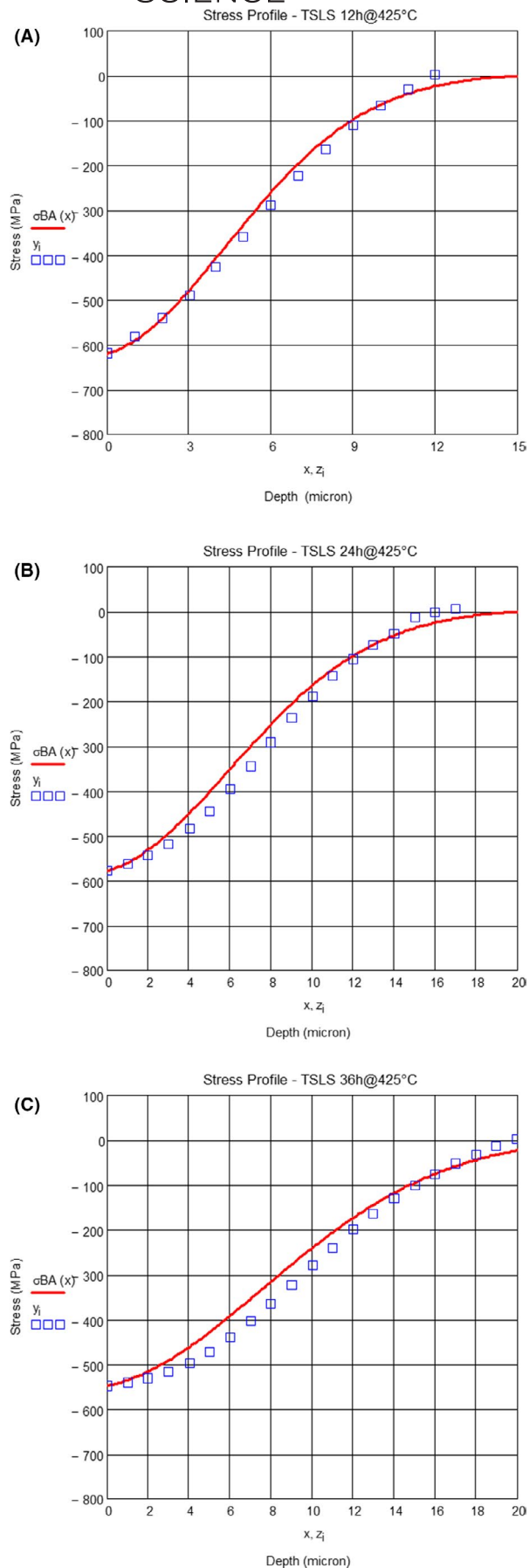


FIGURE 6 A, Comparison between measured and predicted stress profile—Glass TS11 at 425°C, immersion time 12 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. B, Comparison between measured and predicted stress profile—Glass TS11 at 425°C, immersion time 24 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. C, Comparison between measured and predicted stress profile—Glass TS11 at 425°C, immersion time 36 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values [Color figure can be viewed at wileyonlinelibrary.com]

values is limited to a central section of the exchanged volume from about 5 to 20 μm with a maximum difference of about 100 MPa over stress values of 300 MPa. This disagreement is most likely to be addressed to additional dependency of the relaxation time of the relaxation function to the concentration of the incoming ions. As pointed out by Sun and Dugnani,²¹ for soda-lime silicate glass, this dependency may be significant in addressing discrepancies between measured and predicted stress profiles, even though we cannot completely exclude a contribution due to the measurement method assumptions described above. For the sake of this study this disagreement is not critical and we leave its full explanation to further investigations also because it is worth to consider the point that the prediction follows the “sigmoid” shape of residual stress profile for TS11 glass which is not a characteristic of the AS11 glass. In all reported cases both Sc and Cd are predicted within the uncertainty limits declared by the measuring instrument manufacturer. The zero-time surface equi-biaxial stress generated by ion exchange (this is the value before the activation of fast β -relaxation mechanism, that is for $V_0 = 1$) results in 2.34 GPa for TS11 glass and 2.14 GPa for AS11 glass. These values are well below the values calculated in¹¹ for the lithium magnesium aluminosilicate glass of that study which were about 5 GPa. This observation is in agreement with the assumptions of the mathematical model used in this study for the soda-lime silicate (TS11) and the sodium aluminosilicate (AS11) glasses here studied and with the hypothesis of non-relevance, for the studied glasses, of non-isochoric and free-energy-driven relaxation mechanisms triggered by high internal pressure generated by ion exchange.¹¹

6 | CONCLUSIONS

Residual stress profile of two different glass chemical compositions, soda-lime silicate and sodium aluminosilicate, has been experimentally determined by a photoelastic differential refractometric technique for a wide range of temperatures (400°C–450°C) and immersion times (1–36 hours) of ion exchange processes. Experimental

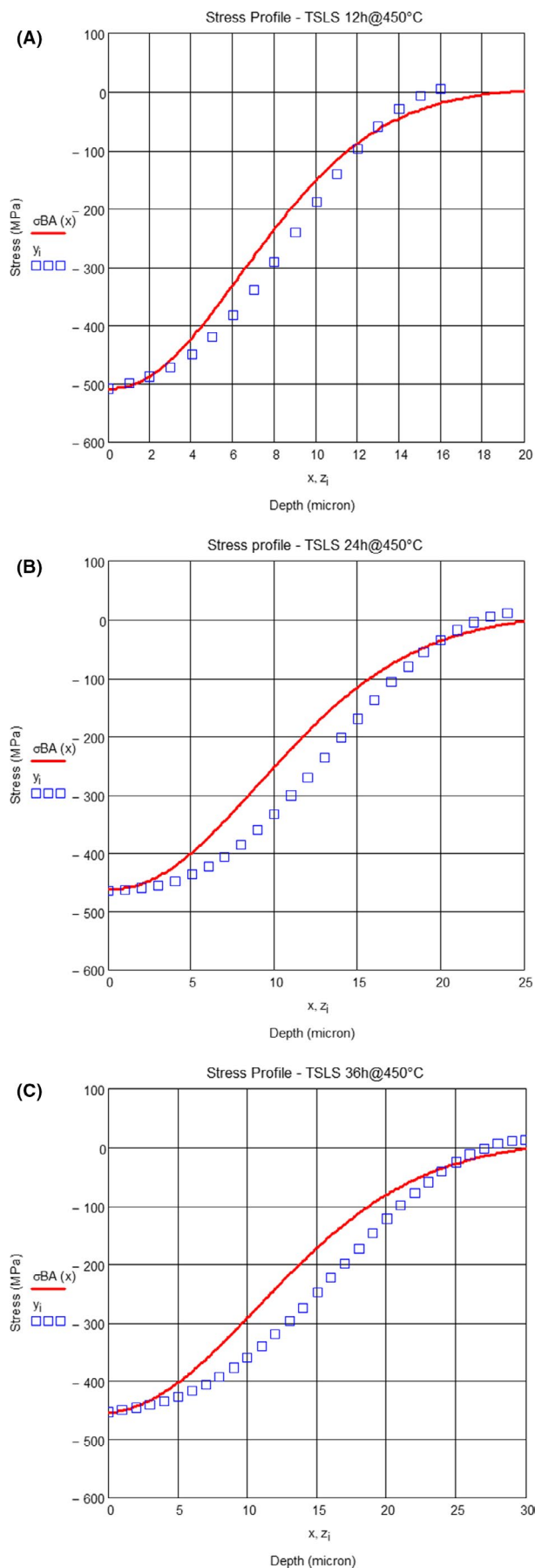


FIGURE 7 A, Comparison between measured and predicted stress profile—Glass TS11 at 450°C, immersion time 12 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. B, Comparison between measured and predicted stress profile—Glass TS11 at 450°C, immersion time 24 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values. C, Comparison between measured and predicted stress profile—Glass TS11 at 450°C, immersion time 36 h. Red line is stress predicted values; Blue boxes $y_i(z_i)$ measured values [Color figure can be viewed at wileyonlinelibrary.com]

values have been compared with predicted ones obtained by a formalized viscoelastic model¹¹ incorporating both stress build up and stress relaxation terms. The comparison between measured and predicted values presents excellent agreement for sodium aluminosilicate glass and similar agreement for soda-lime glass up to a certain temperature and immersion time. A less accurate agreement is noted for soda-lime glass in a limited part of the profile (central area) for the temperature of 450°C and immersion times of 24 and 36 hours. Notwithstanding the recorded disagreement in a central part of the profile for a couple of cases, surface compression and depth of compression values are well predicted for all studied cases. The assumption of relaxation mechanisms in the proposed formalized viscoelastic model for the considered glass compositions (soda-lime and sodium aluminosilicate) is consistent with experimental data and demonstrates different stress relaxation mechanisms which depend on glass chemical compositions. The proposed approach allows accurate residual stress predictions for glasses with modified chemical compositions when submitted to a wide range of ion exchange conditions (immersion time and temperature). Accuracy in residual stress profile determination results in a reliable final strength evaluation through fracture mechanics strength models taking into account interactions between surface flaws geometry and residual stress induced by ion exchange.

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